

TerraGuard – Usability Testing Report

Project Overview

TerraGuard is a disaster awareness and early-warning application UI designed to help users access flood detection information, earthquake alerts, prevention guidance, and emergency onboarding features through a simple user interface.

Project Links

Stitch Prototype: <https://stitch.withgoogle.com/projects/17041649696230127659>

GitHub Repository: <https://github.com/nicolenelson10/terraguard-onboarding-ui>

Screens Designed

- Welcome Screen
- Permissions Screen
- Flood Detection Screen
- Earthquake Prevention Screen
- Success Screen

Objective

Evaluate whether users can easily understand and navigate the onboarding process and access disaster-related information without assistance.

Participants

Five participants between the ages of 18 and 25 tested the prototype.

Test Tasks

1. Open the application.
2. Navigate through the welcome screen.
3. Grant permissions.
4. Access flood detection information.
5. View earthquake prevention guidance.
6. Complete onboarding and reach the success screen.

Positive Feedback

- Users quickly understood the purpose of the application.
- Navigation was simple and intuitive.
- Disaster-related sections were easy to identify.
- The onboarding flow felt smooth.

Issues Found

- Some users were unsure why permissions were required.
- Participants suggested adding a progress indicator.
- Users wanted quick emergency contact options.

Recommendations

- Add explanations before permission requests.
- Include onboarding progress indicators.
- Add emergency help shortcuts.
- Improve visibility of critical alerts.

Results

4 out of 5 users completed all tasks without assistance.
Task Completion Rate: 80%

Conclusion

Usability testing showed that TerraGuard provides a clear and user-friendly onboarding experience. Small improvements in guidance and permissions communication can further enhance usability.